

P8 $A_1, \dots, A_m \subset \{1, \dots, n\}$

$$A_i \not\subset A_j$$

$$a_i = |A_i|$$

$\sigma_1, \dots, \sigma_n$ be random permutation of $1, \dots, n$

$$E_i = \{\{\sigma_1, \dots, \sigma_{a_i}\} = A_i\}$$

$$(\{\sigma_1, \dots, \sigma_n\} = \{1, 2, \dots, n\})$$

Aim: ① $E_1 \cap E_2 = \emptyset$?

② E_1, E_2 indep.?

③ $\sum_{i=1}^m \binom{n}{a_i}^{-1} \ll 1.$

Here $\Omega = \{\text{all permutations}\}$

So $E_i = \{\text{permutations s.t. } \{\sigma_1, \dots, \sigma_{a_i}\} = A_i\}$

① [Contradiction] Suppose that $E_i \cap E_{a_j} \neq \emptyset$.

Then $\exists$ permutation $\sigma_1, \dots, \sigma_n$ s.t. ~~$\{\sigma_1, \dots, \sigma_n\} = \{1, 2, \dots, n\}$~~

$\{\sigma_1, \dots, \sigma_{a_i}\} = A_i$ and $\{\sigma_1, \dots, \sigma_{a_j}\} = A_j$.

we prove more general statement
 ~~$(i \neq j)$~~
 ~~$(i < j)$~~
~~work~~

Since $E_i \cap E_j = \emptyset$, we have $P(E_i \cap E_j) = P(\emptyset) = 0$.

But $P(E_i) \neq 0$ and $P(E_j) \neq 0 \Rightarrow P(E_i)P(E_j) \neq 0$.

(E_i, E_j are nonempty)

$$\therefore P(E_i \cap E_j) \neq P(E_i)P(E_j)$$

$\Rightarrow E_i$ and E_j are dependent.

$$\textcircled{3} \sum_{i=1}^m \binom{n}{a_i}^{-1} \leq 1 \quad \rightarrow \# \text{ ways to assign elements of } A_i \text{ to } 1, 2, \dots, a_i$$

Note that $|E_i| = (a_i)! \cdot (n - a_i)!$

$\rightarrow \# \text{ ways to assign elem. of } \{1, \dots, n\} \setminus A_i$
to $a_i + 1, a_i + 2, \dots, n$

$$\therefore P(E_i) = \frac{(a_i)! (n - a_i)!}{n!} = \frac{1}{\frac{n!}{(a_i)! (n - a_i)!}} = \frac{1}{\binom{n}{a_i}} = \binom{n}{a_i}^{-1}$$

all zeros since they are disjoint

By Excl-Incl. Principle, $P(\cup E_i) = \sum P(E_i) - \sum P(E_i \cap E_j) + \dots$

Note that $P(\cup E_i) \leq 1$. So $\sum P(E_i) \leq 1 \Rightarrow \sum \binom{n}{a_i}^{-1} \leq 1$.